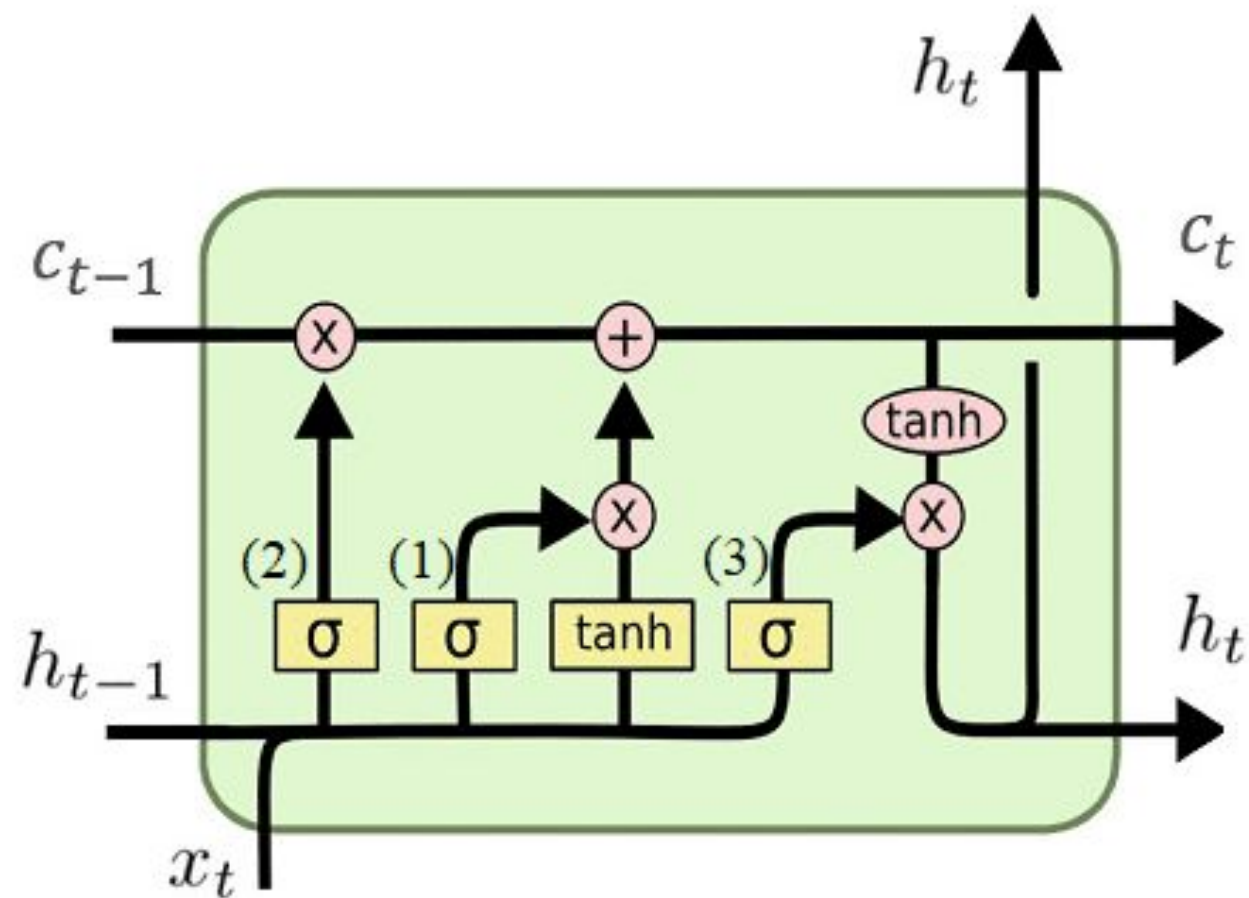


Redes Neuronales Recurrentes (RNNs)

03

Memoria a Largo Plazo (LSTM)

Memoria a largo plazo (LSTM)



Input gate:

$$i_t = \sigma (W^{(ii)}x_t + W^{(hi)}h_{t-1})$$

Forget gate:

$$f_t = \sigma (W^{(if)}x_t + W^{(hf)}h_{t-1})$$

Output gate:

$$o_t = \sigma (W^{(io)}x_t + W^{(ho)}h_{t-1})$$

Process input:

$$\tilde{C}_t = \tanh (W^{(i\tilde{C})}x_t + W^{(h\tilde{C})}h_{t-1})$$

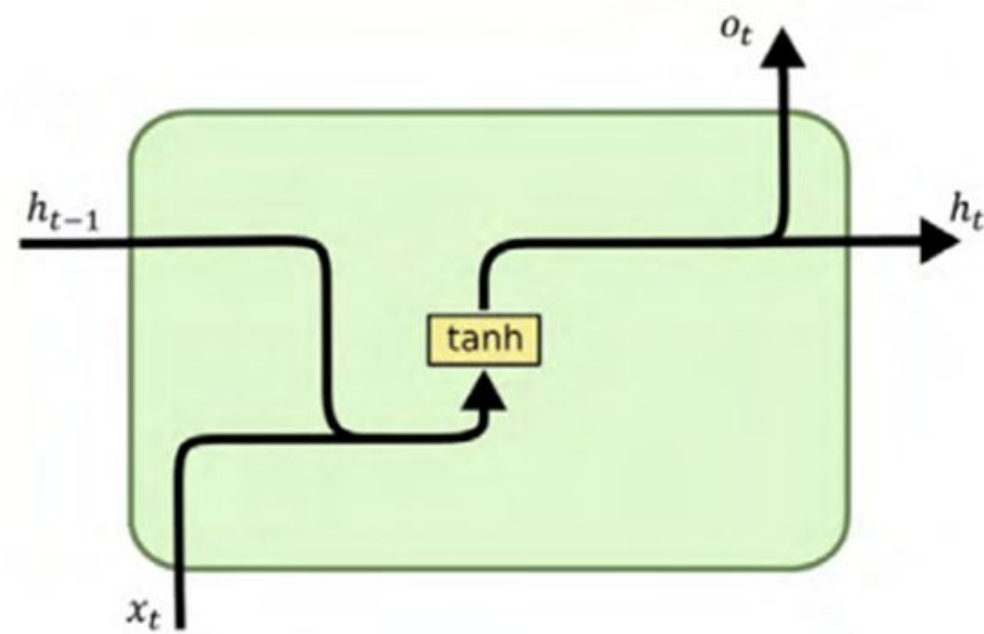
Cell update:

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

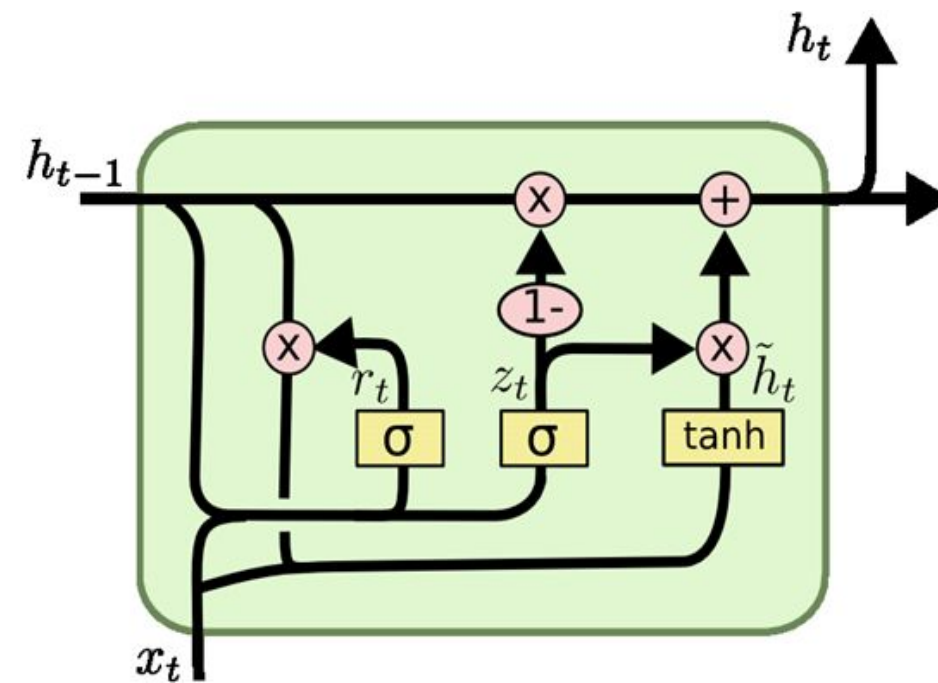
Output:

$$y_t = h_t = o_t * \tanh (C_t)$$

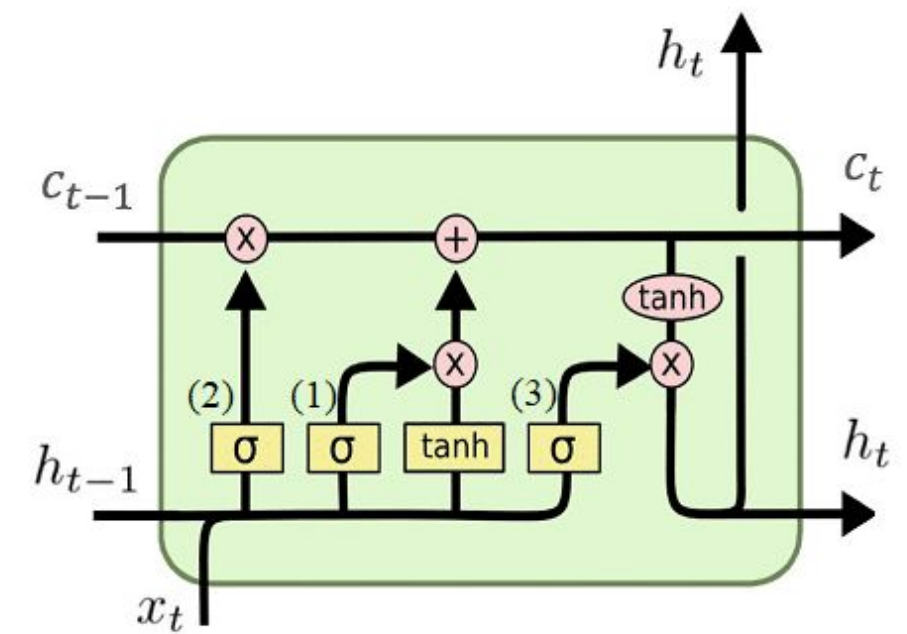
Redes Neuronales Recurrentes



RNN

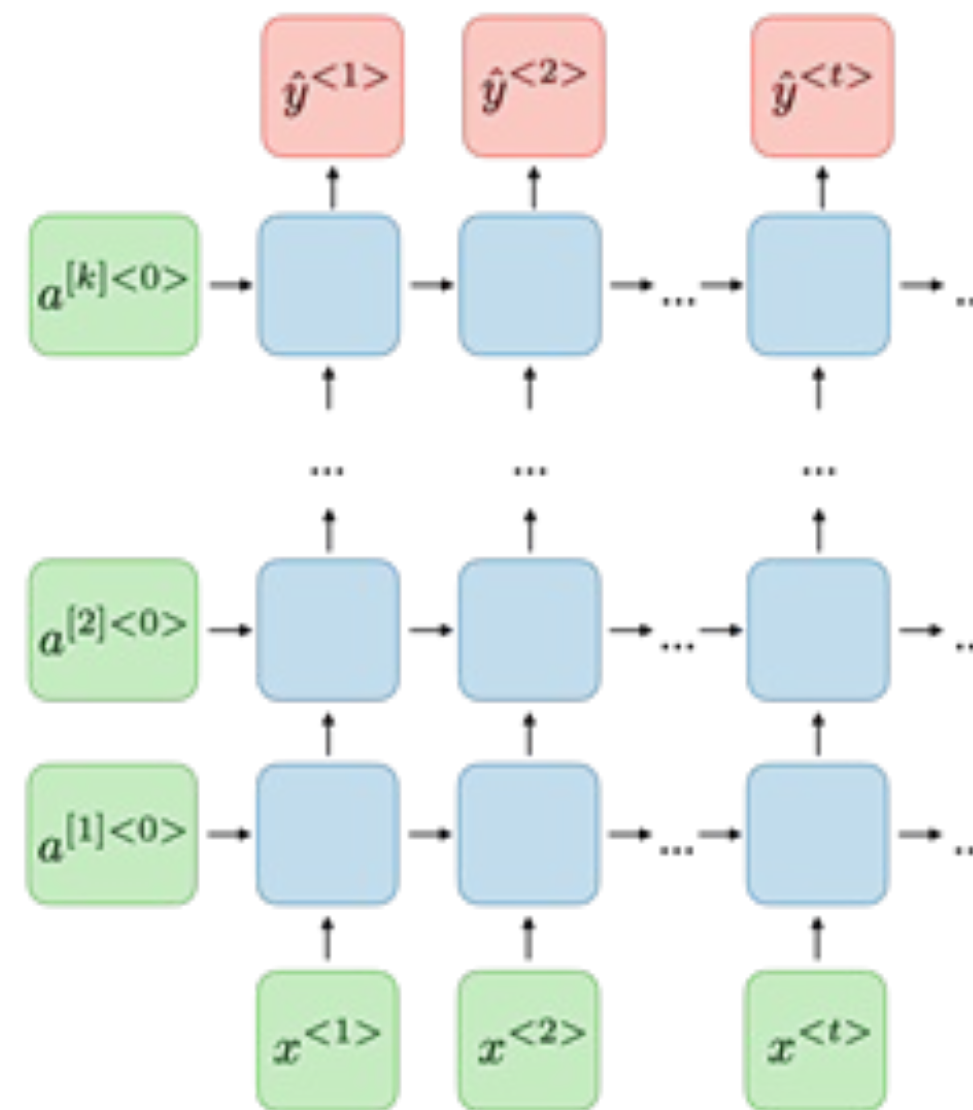
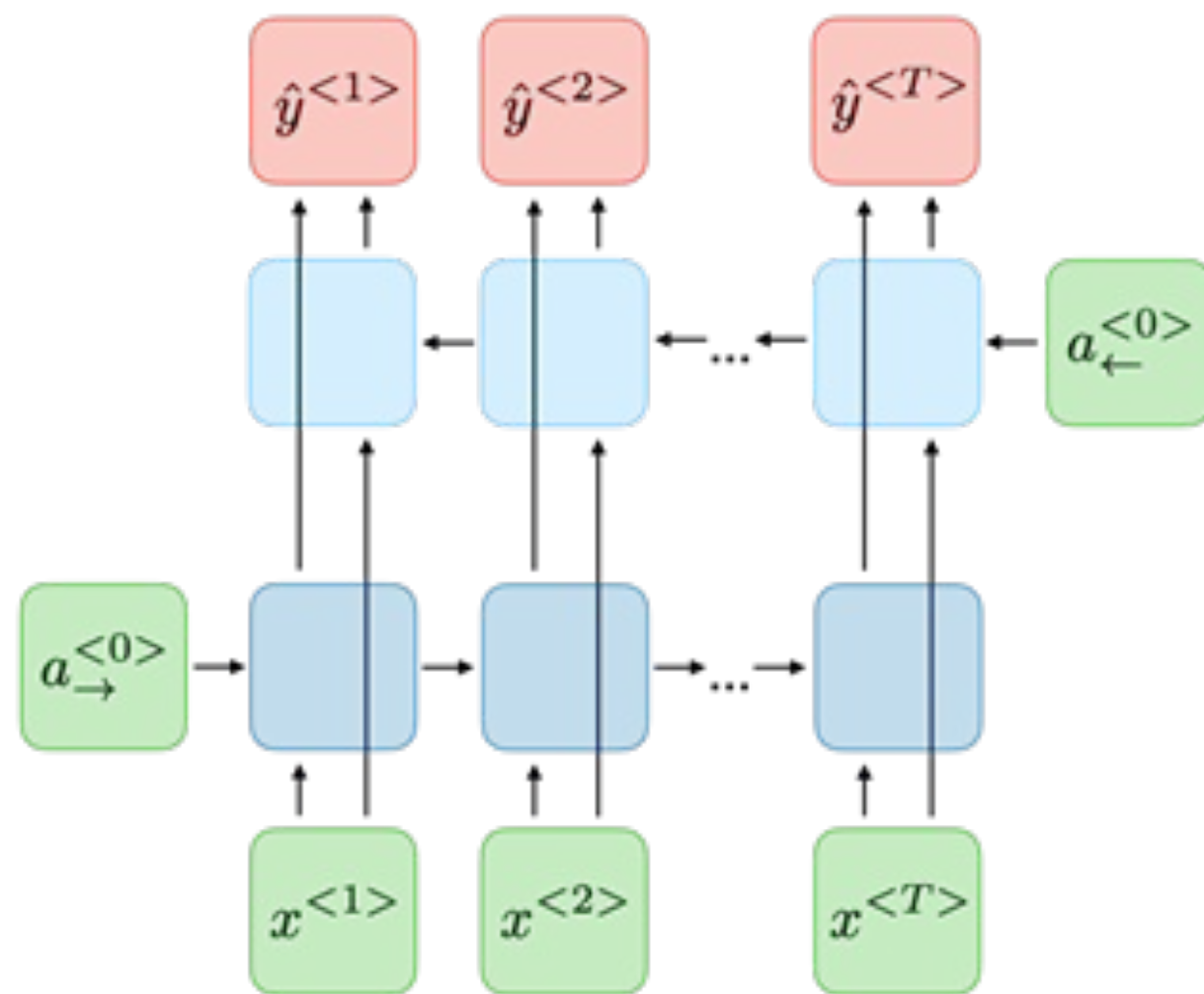


GRU



LSTM

RNN Bidireccionales (BRNN)



¡Muchas gracias!